

A finite absolute-maximizing family almost norms the unit sphere

MathlibAnnex.Satellite.exists_goodSatellitePackage

theorem

Chooses the inverse bound, net and weight before fixing one satellite family.

Statement

Let $n \in \mathbb{N}$ and use reference coordinates $V = \mathbb{R}^n$ with the sup norm. Let $p : V \rightarrow \mathbb{R}$ be a continuous seminorm with specified constants $a_p, b_p > 0$ such that

$$a_p \|x\|_\infty \leq p(x) \leq b_p \|x\|_\infty \quad (x \in V).$$

Thus p is a norm. We use p for the model distance and $\|\cdot\|_\infty$ for reference-coordinate estimates; the two norms are allowed to coincide. A p -dual contraction is a continuous real linear functional r satisfying $|r(x)| \leq p(x)$ for every x . Let $e_1, \dots, e_n$ be the standard coordinate vectors. For an ordered family $F = (\ell_i)_{i=1}^n$ of continuous real linear functionals on V , put

$$M_F = (\ell_i(e_j))_{i,j}, \quad T_F x = (\ell_i(x))_i = M_F x, \quad \mathcal{U}_p = \{F : \forall i, x, |\ell_i(x)| \leq p(x)\}.$$

For a real number η , define the attained maximum and near-maximal set by

$$D_p = \max_{F \in \mathcal{U}_p} |\det M_F| > 0, \quad \mathcal{N}_{p,\eta} = \{F \in \mathcal{U}_p : D_p - \eta \leq |\det M_F|\}.$$

Admissibility is measured with p ; the matrices are written in the fixed reference coordinates. Suppose $0 < \eta < D_p$ and $\varepsilon > 0$. The conclusion chooses a number $K > 0$ with the uniform bound

$$p(M_F^{-1}v) \leq K \|v\|_\infty \quad (F \in \mathcal{N}_{p,\eta}, v \in V),$$

then a finite center set C and a weight $w > 0$. More precisely, put

$$\rho = \frac{\min(\varepsilon, 1)}{8K} > 0, \quad \mathcal{A}_K = \{c \in V : 1/K \leq \|c\|_\infty \leq 1\},$$

$$C \subseteq \mathcal{A}_K, \quad \forall c \in \mathcal{A}_K, \quad \exists a \in C, \quad \|c - a\|_\infty \leq \rho.$$

Index the coefficients by $a \in C$ and put $c_a = a$. For a functional r , let $M_F[i \leftarrow r]$ replace row i by $(r(e_j))_j$. A configuration $Q = (F, s)$ consists of a base frame and a family of additional continuous real linear functionals s_a , called satellites. Its polynomial, admissible set and coefficient budget are

$$\Phi_{w,c}(F, s) = w \det M_F + \sum_{a \in C} \sum_i (c_a)_i \det M_F[i \leftarrow s_a],$$

$$\Omega_p^C = \{(F, s) : F \in \mathcal{U}_p, \forall a \in C, x \in V, |s_a(x)| \leq p(x)\}.$$

$$S_p(c) = D_p \sum_{a \in C} \sum_{i=1}^n |(c_a)_i|.$$

For the selected absolute maximizer $Q = (F, s) \in \Omega_p^C$, that is,

$$|\Phi_{w,c}(Q')| \leq |\Phi_{w,c}(Q)| \quad (Q' \in \Omega_p^C),$$

the same choices satisfy $S_p(c) < w\eta$ and

$$\forall x \in V \ (p(x) = 1), \quad \exists a \in C, \quad 1 - \varepsilon < |s_a(x)|.$$

Assumptions

The real norm model with its comparison constants, $\eta > 0$, $\eta < D_p$ and $\varepsilon > 0$ are the only assumptions. Neither $n > 0$, $\varepsilon < 1$, a nonempty sphere nor nonempty centers are assumed. The number K , its uniform bound, the finite net, and the weight are outputs, not further assumptions.

Conclusion

One finite family, obtained from the selected absolute-maximizing configuration after fixing the net and weight, works for every unit vector. The center/index may depend on the unit vector. Positivity of the weight and its strict budget inequality are also outputs.

Proof route

Fix a uniform inverse bound, choose a finite coefficient net, choose a dominating weight, maximize the absolute polynomial, and then detect each unit vector using that same family.

▼ Proof steps

1. **Choose the common bound before the net.** Apply existence of a common near-maximal inverse bound to the same space V equipped with the norm p and the standard basis and to $0 \leq \eta < D_p$. Its record supplies $K > 0$ and $p(M_F^{-1}c) \leq K\|c\|_\infty$ for every $F \in \mathcal{N}_{p,\eta}$ and c . Set $\rho = \min(\varepsilon, 1)/(8K)$; $\varepsilon > 0$ and $K > 0$ ensure $\rho > 0$.
2. **Take a net in the reference coordinates.** The annulus $\mathcal{A}_K$ is closed and contained in the reference closed unit ball, so it is compact in finite-dimensional V . Apply finite covering of a compact set with this actual set and the nonzero radius ρ (stored as a nonnegative real). Its outputs are a finite $C \subseteq \mathcal{A}_K$ covering every point of $\mathcal{A}_K$ to reference distance at most ρ . The finite coefficient-net record stores exactly its centers, their containment, their finiteness and this cover property. Each $a \in C$ is used as the coefficient vector $c_a = a$; no maximizer or unit vector has yet been chosen.
3. **Turn this net into detection uniformly over near-maximal frames.** For every $F \in \mathcal{N}_{p,\eta}$ and $p(x) = 1$, contraction and reconstruction give

$$\|T_F x\|_\infty \leq 1, \quad 1 = p(x) = p(M_F^{-1}T_F x) \leq K\|T_F x\|_\infty.$$

Thus $T_F x \in \mathcal{A}_K$. Choose $a \in C$ with $\|T_F x - a\|_\infty \leq \rho$. The same inverse estimate gives

$$p(x - M_F^{-1}a) = p(M_F^{-1}(T_F x - a)) \leq K \|T_F x - a\|_\infty \leq K\rho.$$

These are the inputs and output of coefficient detection supplied by this net; detection holds for every near-maximal frame, before the selected maximizer is known.

4. Choose the weight and then the configuration. With the already fixed finite coefficient family, set $w = S_p(c)/\eta + 1$. Nonnegativity of the budget and $\eta > 0$ give $w > 0$ and $S_p(c) < w\eta$. The product Ω_p^C is compact and nonempty: its dual contraction rows are closed, bounded by b_p in reference operator norm, and finite in number; zero rows are admissible. The continuous function $|\Phi_{w,c}|$ therefore attains its maximum. Fix the selected $Q = (F, s)$ once. Apply absolute maximization giving a near-maximal base with this positive weight, strict budget inequality, admissibility and absolute maximum. It gives $F \in \mathcal{N}_{p,\eta}$.

5. Apply detection to the same satellites and retain strictness. For arbitrary $p(x) = 1$, apply the almost-norming satellite estimate with exactly the fixed K, ρ, c, Q , the gap, the detection already proved, and the base membership just obtained. It gives an a from this same family with

$$|s_a(x)| \geq 1 - 2K\rho = 1 - \frac{\min(\varepsilon, 1)}{4} > 1 - \varepsilon.$$

Indeed $0 < \min(\varepsilon, 1) \leq \varepsilon$ implies $\min(\varepsilon, 1)/4 \leq \varepsilon/4 < \varepsilon$. Thus the error is strictly less than ε , without assuming $\varepsilon < 1$. For $n = 0$ the unit sphere is empty, the annulus and centers may be empty, and the final universal statement is vacuous.

A common inverse bound consists of a number $K > 0$ and the uniform estimate

$$p(M_F^{-1}c) \leq K \|c\|_\infty \quad (F \in \mathcal{N}_{p,\eta}, c \in V).$$

It is fixed independently of the particular frame and coefficient. When $\eta < D_p$, every such frame has nonzero determinant. The annulus itself is the coefficient annulus. The strict epsilon version, finite-net absolute-maximizer conclusion and weight choice for a given net preserve this order of choices. As a contextual connection, define $A_Q x = ((\ell_i(x))_i, (s_a(x))_a)$ with the sup norm on its finite coordinates. the upper bound for that coordinate map uses admissibility to give $\|A_Q x\|_\infty \leq p(x)$, while one satellite coordinate bounds the whole map from below gives $1 - \varepsilon < \|A_Q x\|_\infty$ on the unit sphere. For $x \neq 0$, apply the latter to $x/p(x)$ and use linearity to get $(1 - \varepsilon)p(x) < \|A_Q x\|_\infty$, as in the lower bound for every nonzero vector. This only explains the assigned coordinate-map context; no later classification is asserted.

Main citations

- existence of a common near-maximal inverse bound —
MathlibAnnex.DeterminantFrame.nonempty_nearMaxInverseBound
- finite covering of a compact set — Metric.exists_finite_isCover_of_isCompact
- finite coefficient-net record — MathlibAnnex.Satellite.FiniteCoefficientNet
- coefficient detection supplied by this net —
MathlibAnnex.Satellite.finiteCoefficientNet_detectsUnit
- absolute maximization giving a near-maximal base —
MathlibAnnex.Satellite.absoluteMaximizer_base_nearMax
- the almost-norming satellite estimate —
MathlibAnnex.Satellite.absoluteMaximizer_satellites_almost_norm
- the coefficient annulus — MathlibAnnex.Satellite.coefficientAnnulus

- strict epsilon version — `MathlibAnnex.Satellite.absoluteMaximizer_satellites_one_sub_epsilon`
- finite-net absolute-maximizer conclusion — `MathlibAnnex.Satellite.finiteNet_absoluteMaximizer_satellites_one_sub_epsilon`
- weight choice for a given net — `MathlibAnnex.Satellite.exists_goodNetSatelliteMaximizer`
- the upper bound for that coordinate map — `MathlibAnnex.FiniteSup.configurationMap_norm_le_model`
- one satellite coordinate bounds the whole map from below — `MathlibAnnex.FiniteSup.configurationMap_norm_gt_of_satellite`
- the lower bound for every nonzero vector — `MathlibAnnex.FiniteSup.maxNetConfigurationMap_lower_of_ne_zero`
- Exact declaration and proof — `MathlibAnnex.Satellite.exists_goodSatellitePackage`

Lean source signature (exact)

```
theorem exists_goodSatellitePackage
  {n : ℕ} (M : NormModel n) {η ε : ℝ}
  (hη : 0 < η) (hηD : η < detMax M) (hε : 0 < ε) :
  ∃ H : NearMaxInverseBound M η,
    ∃ C : FiniteCoefficientNet (n := n) H.boundConstant
      (satelliteRadiusNNReal ε H.boundConstant hε H.boundConstant_pos),
      ∃ weight : ℝ,
        0 < weight ∧
        satelliteBudget M (Internal.centerValue C) < weight * η ∧
        ∀ x : Coord n, M.p x = 1 →
          ∃ a : C.centers,
            1 - ε <
              |(maxSatelliteConfiguration M C.centers
                weight (Internal.centerValue C)).2 a x|
```

The nested existence quantifiers choose H , then the finite net C , then weight . The final assertion tests the resulting one family on every unit vector.

In the source	Mathematical meaning
<code>Coord n</code>	The reference space $V = \mathbb{R}^n$ with its sup norm; <code>Coord n</code> abbreviates $\text{Fin } n \rightarrow \mathbb{R}$. Lean uses indices $0, \dots, n-1$ and the formulas use $1, \dots, n$.
<code>M : NormModel n; M.p</code>	The input M contains the continuous seminorm p and constants $a_p, b_p > 0$ with $a_p \ x\ _\infty \leq p(x) \leq b_p \ x\ _\infty$ for every x . $M.p$ x is the scalar $p(x)$.
<code>detMax M; nearMaxFrames M η</code>	$D_p = \max_{F \in \mathcal{U}_p} \det M_F $ and $\mathcal{N}_{p,\eta} = \{F \in \mathcal{U}_p : D_p - \eta \leq \det M_F \}$, where $\mathcal{U}_p = \{(\ell_i) : \forall i, x, \ell_i(x) \leq p(x)\}$.
<code>hη; hηD; hε</code>	The only numerical input hypotheses are $0 < \eta, \eta < D_p$, and $0 < \varepsilon$.

In the source	Mathematical meaning
$\exists H : \text{NearMaxInverseBound } M \ \eta$	There exists $K > 0$ with $p(M_F^{-1}v) \leq K\ v\ _\infty$ for all $F \in \mathcal{N}_{p,\eta}$ and $v \in V$. The structure H stores that number and its two stated properties.
$H.\text{boundConstant}; H.\text{boundConstant_pos}$	The chosen number K and the proof that $K > 0$.
$\text{satelliteRadiusNNReal } \varepsilon \ H.\text{boundConstant} \ \text{he} \ H.\text{boundConstant_pos}$	The radius $\rho = \min(\varepsilon, 1)/(8K) > 0$, represented as a nonnegative real number. The proof arguments establish the required sign; they are not further numerical choices.
$\exists C : \text{FiniteCoefficientNet } (n := n) \ H.\text{boundConstant} \ (\dots)$	There exists a finite set $C \subseteq \mathcal{A}_K = \{v : 1/K \leq \ v\ _\infty \leq 1\}$ such that every $v \in \mathcal{A}_K$ has $a \in C$ with $\ v - a\ _\infty \leq \rho$. The structure stores the set, containment, finiteness and cover property.
$C.\text{centers}; a : C.\text{centers}; \text{Internal.centerValue } C \ a$	The center set, a center together with its membership proof, and its coordinate vector $a \in V$, respectively. The coefficient is $c_a = a$.
$\text{satelliteBudget } M \ (\text{Internal.centerValue } C)$	$S_p(c) = D_p \sum_{a \in C} \sum_i a_i $, using this chosen finite center set.
$\exists \text{weight} : \mathbb{R}, \ 0 < \text{weight} \wedge \text{satelliteBudget } M \ (\dots) < \text{weight} * \eta$	After C is chosen there exists a real w satisfying both $w > 0$ and $S_p(c) < w\eta$. These are output properties.
$\text{maxSatelliteConfiguration } M \ C.\text{centers} \ \text{weight} \ (\text{Internal.centerValue } C)$	The selected pair $Q = (F, s) \in \Omega_p^C$ maximizing $ \Phi_{w,c} $ for this p, C, w, c . Every base row and every s_a is dominated by p ; the polynomial is the displayed weighted determinant plus row-replacement sum.

In the source	Mathematical meaning
$(\text{maxSatelliteConfiguration } M \ C.\text{centers} \ \text{weight} \ (\text{Internal.centerValue } C)).2 \ a \ x$	The entire expression is $s_a(x)$. .2 selects the satellite family in the one selected pair, a selects its functional, and x is the evaluation point.
$\forall x : \text{Coord } n, \ M.p \ x = 1 \rightarrow \exists a : C.\text{centers}, \ 1 - \varepsilon < \dots $	For every $x \in V$ with $p(x) = 1$, some index a from that same finite set satisfies $1 - \varepsilon < s_a(x) $. The family is fixed before x ; only a may depend on x .

$\wedge$ means “and” and the implication $M.p \ x = 1 \rightarrow \dots$ restricts the final assertion to unit vectors. $(n := n)$ fixes the dimension of the net. The net structure and its center set are distinct; the source writes them as C and $C.\text{centers}$, while the formulas use $\mathcal{C}$ for the set.

▼ Relevant surrounding context (separate exact excerpts)

Exact source lines 13–14:

```
private abbrev Coord (n : ℕ) := Fin n → ℝ
private abbrev NormModel (n : ℕ) := EquivalentSeminorm (Coord n)
```

Exact source lines 54–54:

```
private abbrev NearMaxInverseBound {n : ℕ} (M : NormModel n) := DeterminantFrame.NearMaxInverseBound
(modelBasis M)
```

Full surrounding source. These excerpts are separate from the declaration above.

Surrounding assumptions and aliases: exact source, lines 7–55. The full original context is retained with the source evidence.

▼ **Exact content identity**

Declaration: MathlibAnnex.Satellite.exists_goodSatellitePackage

Accepted content SHA-256: f79f8ef5f24914391336ebe540e799da4107cab5645e0372791ac71c98ea6b32

Accepted source guide SHA-256: dc8964fc07f86e1508cc438b70e4954d4c7c61ded2485a99e6ae441e39d7e78b

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73